# D13 | XOR decision block

All-case aligned observations: 41,333 | Train: 28,952 | Held-out test: 12,381

Test accuracy 0.890 | Majority baseline 0.890 | Balanced accuracy 0.500 | Macro-F1 0.471

B1 -> M09 | aligned support 4,548
B2 -> M12 | aligned support 36,785

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 594.5 AND c.AppType != "Limit raise" AND e.FirstWithdrawal <= 20.825 -> B2 [train n=4,961]

e.CreditScore <= 594.5 AND c.AppType != "Limit raise" AND e.FirstWithdrawal > 20.825 -> B2 [train n=11,801]

e.CreditScore <= 594.5 AND c.AppType = "Limit raise" AND e.FirstWithdrawal <= 1508 -> B2 [train n=1,815]

e.CreditScore <= 594.5 AND c.AppType = "Limit raise" AND e.FirstWithdrawal > 1508 -> B2 [train n=441]

e.CreditScore > 594.5 AND c.AppType != "New credit" AND c.RequestedAmount <= 2650 -> B2 [train n=89]

e.CreditScore > 594.5 AND c.AppType != "New credit" AND c.RequestedAmount > 2650 -> B2 [train n=249]

e.CreditScore > 594.5 AND c.AppType = "New credit" AND c.LoanGoal != "Remaining debt home" -> B2 [train n=9,267]

e.CreditScore > 594.5 AND c.AppType = "New credit" AND c.LoanGoal = "Remaining debt home" -> B2 [train n=329]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.